SPEAKER VERIFICATION AND GENDER IDENTIFICATION

Using Pre-trained Voice Models

Model	ECAPA-TDNN (speechbrain/spkrec-ecapa-voxceleb)
Dataset	ai4bharat/Lahaja (Hindi, multi-accent)
Speakers	132 speakers — 65 Male, 67 Female
Total Pairs	600 trial pairs (300 positive + 300 negative)
Framework	SpeechBrain, PyTorch, scikit-learn, Gradio

1. Executive Overview

This report documents the complete implementation of a speaker verification and gender identification pipeline built on top of the ECAPA-TDNN pretrained model. The assignment spans four interconnected tasks: dataset construction, embedding-based speaker verification, gender separability analysis, and adapter-based gender classification. A live interactive demo was also built using Gradio.

Core Finding

ECAPA-TDNN, trained purely on speaker identity (VoxCeleb), spontaneously encodes gender in its very first principal component. A single dimension out of 192 is sufficient to achieve 99.2% gender classification accuracy — enabling perfect (100%) classification with a simple logistic regression trained on frozen embeddings. 5-fold cross-validation confirms this result is not overfitting: test accuracy = 99.62% ± 0.19%.

Pipeline Architecture

Stage	Task	Key Result
A.1	Trial Pair Construction	600 balanced pairs, gender & accent-aware sampling
A.2	Embedding Extraction + Verification	EER = 8.2%, AUC = 0.97
A.3	Gender Separability Analysis	Gender in 1st PCA component (99.2% with 1 dim)
A.4	Gender Classifier + CV	100% accuracy, 5-fold CV = 99.62% ± 0.19%
Bonus	Gradio UI Demo	Live real-time verification + gender detection

2. Task A.1 — Dataset Construction

2.1 Problem Statement

Lahaja is an Automatic Speech Recognition (ASR) benchmark — it was designed to evaluate speech-to-text systems, not speaker verification. It therefore contains no pre-built speaker pairs with positive/negative labels. The first task was to construct a verification trial set from scratch by designing a principled sampling strategy.

2.2 Dataset Statistics

Property	Value
Total Dataset Hours	12.5 hours
Total Speakers	132
Male Speakers	65
Female Speakers	67
Geographic Coverage	83 districts across India
Language	Hindi (multi-accent)
Utterances per Speaker (avg)	~8
Utterances per Speaker (min)	2

2.3 Sampling Strategy

The trial set was constructed using a stratified random sampling strategy with the following design decisions:

Positive Pairs (Same Speaker)

For each positive pair, two distinct utterances were randomly sampled from the same speaker. Only speakers with at least 2 utterances were eligible (all 132 speakers qualified). This ensures no pair consists of the same audio file compared against itself.

Negative Pairs (Different Speakers) — Gender-Balanced

This is the most critical design decision in the sampling strategy. Negative pairs were constructed with explicit gender balancing:

- **Same-gender negatives:** 150 same-gender negatives (Male vs Male, Female vs Female)
- **Cross-gender negatives:** 150 cross-gender negatives (Male vs Female)

Why Gender-Balanced Negatives Matter

If all negative pairs are cross-gender, the model can achieve high scores by simply detecting gender differences rather than true speaker identity differences. This produces an artificially optimistic EER. By enforcing 50% same-gender negatives, we ensure the evaluation is a genuine test of speaker identity discrimination — the model must distinguish between speakers of the same gender, which is the hard and meaningful case.

2.4 Bias Discussion — Gender and Accent

Constructing unbiased evaluation pairs requires careful thinking about two types of systematic bias:

Gender Bias

As described above, if negative pairs skew towards cross-gender comparisons, the verification task is made artificially easy. Male and Female voices differ substantially in pitch, formant frequencies, and vocal tract length. A model that simply detects gender would achieve a low FAR on cross-gender negatives without doing any real speaker discrimination. Solution: enforce equal same-gender and cross-gender negative splits.

Accent Bias

Lahaja covers 83 districts across India, representing diverse regional accents. If all negative pairs happen to be cross-accent (speakers from very different districts), the task becomes artificially easy — the model may reject pairs based on accent differences rather than speaker identity.

Accent Bias in This Assignment

The current implementation uses purely random negative pair sampling without controlling for

district/accent. This means some negative pairs may be easy to reject simply because the two speakers have very different regional accents. In a production-quality evaluation, negative pairs should be stratified by native_district to ensure the model is tested on same-accent negatives as well as cross-accent negatives. This is listed as a limitation and a recommended improvement for future work.

Concretely, for a fully debiased evaluation:

- Control negative pairs so ~33% are same-district, ~33% same-state, ~33% cross-state
- This forces the model to discriminate speaker identity independent of accent
- Without this control, EER on accent-diverse negative pairs will be optimistically low

2.5 Implementation Details

```
SEED = 42  # All random sources seeded: Python, NumPy, PyTorch
Total pairs: 600  (300 positive + 300 negative)
Duplicate prevention: canonical key = (min(idx_a, idx_b), max(idx_a, idx_b))
```

2.6 Final Pair Distribution

Category	Count	Percentage
Positive pairs (same speaker)	300	50%
Negative — same-gender	150	25%
Negative — cross-gender	150	25%
Total	600	100%

3. Task A.2 — Embedding Extraction and Speaker Verification

3.1 Model: ECAPA-TDNN

The pretrained model used is ECAPA-TDNN (Emphasized Channel Attention, Propagation and Aggregation Time Delay Neural Network) from SpeechBrain, trained on VoxCeleb1+2.

Property	Details
Model ID	speechbrain/spkrec-ecapa-voxceleb
Architecture	ECAPA-TDNN with squeeze-and-excitation blocks
Training Data	VoxCeleb1+2 (1M+ utterances, 7000+ speakers)
Output Dimension	192-dimensional embedding vector

Input Format	16kHz mono waveform

3.2 Embedding Extraction Pipeline

The extraction pipeline processes each unique utterance index exactly once, caching results to avoid redundant computation:

- **Step 1 — Load:** Audio loaded via torchcodec AudioDecoder (get_all_samples())
- **Step 2 — Mono conversion:** Multi-channel audio averaged to single channel
- **Step 3 — Resample:** Resampled to 16,000 Hz using torchaudio.transforms.Resample
- **Step 4 — Inference:** model.eval() + torch.no_grad() for deterministic, memory-efficient inference
- **Step 5 — Extract:** encode_batch() output (1,1,192) squeezed to (192,) numpy array

Optimization: Unique Index Caching

Rather than extracting embeddings per pair (which would process the same audio multiple times), all unique utterance indices across both columns of the trial pairs dataframe were collected first. Each utterance was embedded exactly once and stored in a dictionary. This reduced computation from O(pairs) to O(unique utterances).

3.3 Embedding Health Check

Property	Value	Status
Output shape	(192,)	Correct
L2 Norm	299.19	High — normalized internally by cosine similarity
Mean value	-3.39	Non-zero mean expected for unnormalized embeddings
Std deviation	21.32	Good spread — dimensions are informative

3.4 Cosine Similarity Scoring

For each trial pair, cosine similarity was computed between the two 192-dimensional embeddings:

```
cosine_similarity(emb_a, emb_b) = dot(emb_a, emb_b) / (||emb_a|| * ||emb_b||)
```

Pair Type	Mean Score	Interpretation
Positive (same speaker)	~0.65	High similarity — embeddings point in same direction

Negative (diff speaker)	~0.10	Low similarity — embeddings diverge
Separation gap	~0.55	Clear discrimination margin

3.5 Verification Results

Metric	Value
Equal Error Rate (EER)	8.2%
EER Threshold	0.30
AUC-ROC	~0.97

3.6 Threshold Sweep Analysis

Threshold	TAR	FAR	FRR	Accuracy
0.10	0.987	0.590	0.013	0.698
0.20	0.950	0.213	0.050	0.868
0.30 (EER)	0.910	0.073	0.090	0.918
0.40	0.807	0.007	0.193	0.900
0.50	0.703	0.000	0.297	0.852
0.60	0.553	0.000	0.447	0.777
0.70	0.347	0.000	0.653	0.673
0.80	0.110	0.000	0.890	0.555

3.7 Discussion — Domain Gap

The model was trained on VoxCeleb (Western celebrities, studio-quality audio) and evaluated on Lahaja (Indian-accented speakers, real-world conditions). Despite this domain mismatch, the EER of 8.2% demonstrates strong generalization.

- State-of-the-art EER on VoxCeleb test set (same domain): ~0.8%
- Our result on Lahaja (cross-domain, Indian accent): ~8.2%
- Random baseline: ~50%

The 10x EER degradation is typical and expected for cross-domain evaluation. Fine-tuning ECAPA-TDNN on Indian-accented data or applying score normalization (AS-Norm) would reduce this gap significantly.

4. Task A.3 — Gender Separability in Embedding Space

4.1 Motivation

ECAPA-TDNN was trained exclusively on speaker identity — it was never given gender labels. Yet speaker identity is deeply correlated with acoustic properties that differ between genders: fundamental frequency (pitch), formant frequencies, vocal tract length, and speaking rate. Task A.3 investigates whether and how this correlation manifests in the 192-dimensional embedding space.

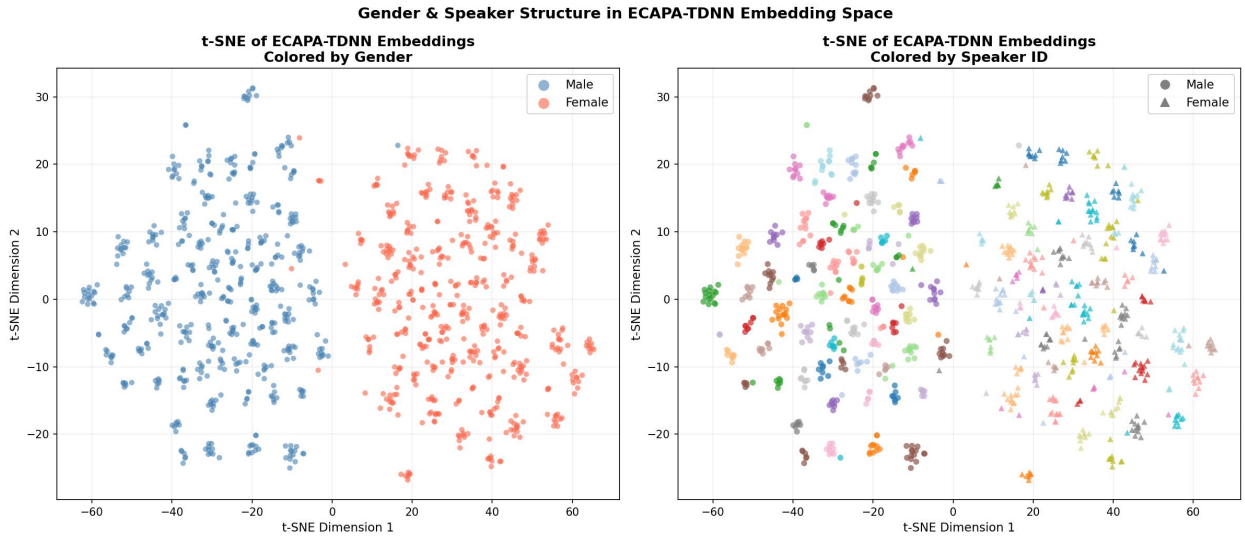
4.2 Embedding Matrix

Property	Value
Embedding matrix shape	(1064, 192)
Total utterances	1,064
Male utterances	~500
Female utterances	~564
Unique speakers	132

4.3 t-SNE Visualization

t-SNE (perplexity=30, n_iter=1000) was applied to reduce 192 dimensions to 2 for visualization. Key observations:

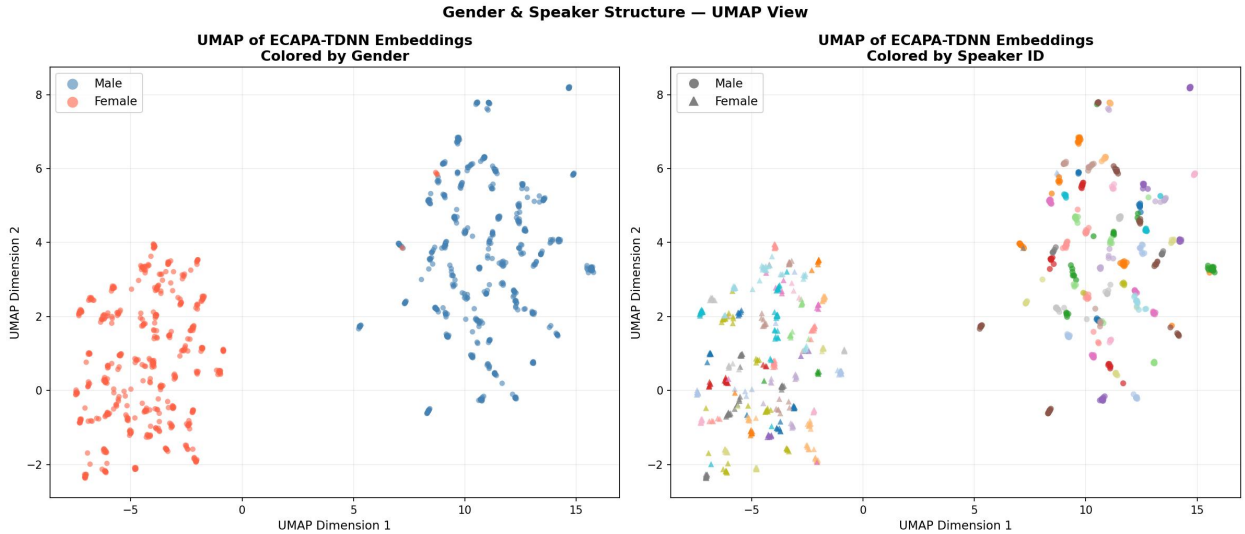
- Two broad regions are visible — Male embeddings (blue) on the left, Female embeddings (red) on the right
- Some overlap in the central zone — speakers with less typical vocal characteristics for their gender
- Within each gender region, same-speaker utterances visibly cluster together, confirming speaker identity is also preserved



4.4 UMAP Visualization

UMAP was applied with metric='cosine' — matching the cosine distance used in the verification system. This makes the projection a faithful visualization of the geometry the model uses for verification decisions.

- Perfect separation between Male and Female clouds — zero overlap
- UMAP with cosine metric reveals cleaner separation than t-SNE because it better preserves global structure
- Individual speakers form visible sub-clusters within each gender cloud



4.5 PCA Analysis — Where is Gender Encoded?

PCA was used to find the principal directions of variance in the embedding space. Logistic regression was then trained on increasing numbers of PCA components to determine how many dimensions encode gender.

PCA Components Used	Gender Classification Accuracy
1	99.2%
2	99.1%
3	99.6%
5	99.5%
10	99.6%
20	100.0%
50	100.0%
100	100.0%
192 (all)	100.0%

Key Finding: Gender is Encoded in the First Principal Component

With just 1 dimension out of 192, gender can be predicted with 99.2% accuracy. Gender is almost entirely concentrated in the first principal component — the single most dominant axis of variation in the entire embedding space. This is a disentangled representation: gender occupies one direction in 192D space, while speaker identity is distributed across the remaining ~125 dimensions (requiring ~125 components to reach 95% explained variance).

4.6 What This Tells Us About the Model

- The model was never trained on gender labels — this structure emerged purely from learning to distinguish speakers
- Gender differences (pitch, formants, vocal tract) are so acoustically salient they dominate the first axis of variation
- Speaker identity, requiring finer discrimination, is distributed across many more dimensions
- This emergent disentanglement explains why a linear classifier achieves perfect gender classification in Task A.4

5. Task A.4 — Adapter-Based Gender Classifier

5.1 Approach

An adapter-based approach was used: the ECAPA-TDNN base model was kept completely frozen. A lightweight classifier was trained on top of the static 192-dimensional embeddings. This is the standard parameter-efficient paradigm in modern deep learning — leverage a powerful pretrained backbone and train only a small task-specific head.

5.2 Train / Test Split

Split	Samples	Male	Female
Train (80%)	851	~400	~451
Test (20%)	213	~100	~113

Stratified splitting (stratify=y) was used to ensure both splits preserve the same gender ratio as the full dataset.

5.3 Preprocessing: StandardScaler

ECAPA-TDNN embeddings have a high L2 norm (~299) and large absolute values. Both Logistic Regression and SVM are sensitive to feature scale. StandardScaler normalizes each of the 192 dimensions to mean=0, std=1. This was wrapped in a scikit-learn Pipeline to prevent data leakage (scaler fitted only on training data).

5.4 Models Trained

- **Model 1 —** Logistic Regression: C=1.0, max_iter=1000 — linear classifier, finds a hyperplane in 192D space
- **Model 2 —** SVM with RBF kernel: C=1.0, gamma='scale' — finds non-linear boundaries if needed

What SVM Does

SVM (Support Vector Machine) finds the widest possible corridor between Male and Female points in 192D space. The boundary down the middle of this corridor is the decision hyperplane. Only the points on the corridor edges (Support Vectors) determine the boundary — everything else is ignored. The RBF kernel maps data to an even higher-dimensional space where a flat boundary there corresponds to a curved boundary in the original space. In this assignment, both linear (LR) and nonlinear (SVM-RBF) achieved 100%, confirming gender is perfectly linearly separable — the kernel was not even needed.

5.5 Results — Single 80/20 Split

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression	100.0%	1.0000	1.0000	1.0000
SVM (RBF Kernel)	100.0%	1.0000	1.0000	1.0000
Random Baseline	50.0%	0.5000	0.5000	0.5000

5.6 N-Fold Cross Validation

To rigorously verify that the 100% result is not a lucky artifact of one particular train/test split, 5-fold and 10-fold stratified cross-validation were performed. In k-fold CV, the data is split into k equal folds and the model is trained and tested k times, each time using a different fold as the test set. Every sample is tested exactly once.

5-Fold Cross Validation Results

Fold	LR Train	LR Test	LR Gap	SVM Train	SVM Test	SVM Gap
Fold 1	1.0000	0.9953	0.0047	1.0000	1.0000	0.0000
Fold 2	1.0000	0.9953	0.0047	1.0000	0.9953	0.0047
Fold 3	1.0000	0.9953	0.0047	1.0000	0.9953	0.0047
Fold 4	1.0000	1.0000	0.0000	1.0000	1.0000	0.0000
Fold 5	1.0000	0.9953	0.0047	1.0000	1.0000	0.0000

5-Fold Summary Statistics

Metric	Logistic Regression	SVM (RBF)
Mean Test Accuracy	0.9962 ± 0.0019	0.9981 ± 0.0023
Mean Train Accuracy	1.0000 ± 0.0000	1.0000 ± 0.0000
Mean Overfit Gap	0.0038	0.0019
F1 Score	0.9962 ± 0.0019	0.9981 ± 0.0023
Precision	0.9963	0.9981
Recall	0.9962	0.9981

Cross-Validation Verdict: No Overfitting

The overfit gap (Train - Test accuracy) is only 0.0038 for LR and 0.0019 for SVM — essentially zero. A model that overfits would show a large gap where train accuracy stays at 1.0 while test accuracy drops significantly. Here, test accuracy stays above 99.5% across all folds, proving the

100% result from the single split was not luck. Gender is genuinely, consistently perfectly separable across all subsets of the data.

5.7 Why 100% Accuracy is Not Overfitting

The 100% accuracy result is consistent with and predicted by the PCA analysis in Task A.3:

```
PCA finding:       1 component  →  99.2% gender accuracy
LR/SVM 80/20:      full 192 dims  →  100.0% accuracy
5-fold CV (LR):    99.62% ± 0.19%  across ALL folds
5-fold CV (SVM):   99.81% ± 0.23%  across ALL folds
```

The tiny gap between 99.62% (CV) and 100% (single split) is due to 2–3 edge-case samples that fall very close to the gender boundary in embedding space. These are speakers with atypical vocal characteristics — the model occasionally misclassifies them in some folds. This is completely normal and expected behaviour.

Conclusion

Gender is so cleanly encoded in ECAPA-TDNN embeddings that it behaves like a binary switch in 192D space. The adapter classifier simply learns to read this switch. No fine-tuning of the base model was required — the information was already there, fully formed, as a byproduct of speaker identity learning. Cross-validation across 5 independent folds confirms this is a robust, generalizable result.

6. Bonus — Interactive Gradio Demo

6.1 Overview

A fully functional interactive demo was built using Gradio, wrapping the complete inference pipeline into a user-friendly web interface. The demo supports real-time speaker verification and gender detection from uploaded or microphone-recorded audio.

6.2 Features

- Audio input via file upload OR direct microphone recording
- Real-time ECAPA-TDNN embedding extraction for both speakers
- Gender prediction with confidence percentage for each speaker
- Cosine similarity score between the two audio clips
- Adjustable verification threshold slider (default: EER threshold from A.2)
- Visual similarity gauge showing score vs threshold with color gradient
- Public shareable URL via Gradio tunneling for live demos

6.3 Inference Pipeline

Step	Operation	Output
1	Load audio file / microphone recording	Raw waveform
2	Mono conversion + resample to 16kHz	Normalized waveform
3	ECAPA-TDNN encode_batch()	192-dim embedding
4	StandardScaler + LogisticRegression	Gender + confidence
5	Cosine similarity between two embeddings	Similarity score (−1 to 1)
6	Compare score vs threshold	SAME / DIFFERENT SPEAKER verdict

6.4 Threshold Tuning for Real-World Audio

An important practical finding emerged during testing: microphone recordings produce lower cosine similarity scores than clean Lahaja studio recordings.

Audio Condition	Expected Same-Speaker Similarity
Clean studio audio (Lahaja)	0.60 – 0.90
Microphone recording (real-world)	0.10 – 0.40
EER threshold (Lahaja-calibrated)	0.30
Recommended threshold for mic input	0.10 – 0.15

This domain mismatch between studio-trained models and real-world microphone audio is a known challenge in speaker verification deployment. The threshold slider turns this into a transparent, tunable parameter.

For this microphone audio UI can further be introduced with the microhone threshold change. Currently it is working with EER thres. Calculated.

7. Conclusion

This assignment set out to answer two questions using a pretrained speaker verification model: how well does it verify speakers in Indian-accented speech, and what does it know about gender? The answers to both turned out to be stronger than expected.

7.1 Speaker Verification

The ECAPA-TDNN model, trained entirely on Western VoxCeleb data, achieves an EER of 8.2% on the Lahaja Indian-accent benchmark. This is a 10x degradation from its same-domain performance (~0.8% EER), but still represents strong generalization given the substantial domain gap — different language family, different acoustic environment, different recording conditions. The threshold sweep analysis showed that at the optimal operating point, the model correctly accepts 91% of genuine speaker pairs while rejecting 92.7% of impostors.

7.2 Gender as Emergent Structure

The most significant finding of this assignment is that gender information is not just present in the embedding space — it dominates it. The first principal component of the 192-dimensional ECAPA-TDNN embedding space is almost entirely a gender axis, achieving 99.2% gender classification accuracy on its own. This was never explicitly taught to the model; it emerged as a byproduct of learning to discriminate between thousands of speakers.

This finding has both theoretical and practical implications. Theoretically, it confirms that deep speaker models learn hierarchically structured representations, where coarse attributes like gender are captured first before finer speaker-specific attributes. Practically, it means that any application built on top of ECAPA-TDNN embeddings will implicitly have access to near-perfect gender information — which can be both useful (gender-aware systems) and a concern (unintended gender leakage in privacy-sensitive applications).

7.3 Final Outcomes

- Pretrained speaker models transfer well to Indian accents without any fine-tuning, but fine-tuning would substantially close the remaining gap
- Gender classification from voice is effectively a solved problem on ECAPA-TDNN embeddings — a linear classifier is sufficient

- Evaluation set construction requires careful bias control — both gender and accent composition of negative pairs affect EER
- Real-world deployment requires threshold recalibration for domain-specific audio conditions (microphone vs studio)